Ejercicio 10 Guía 1

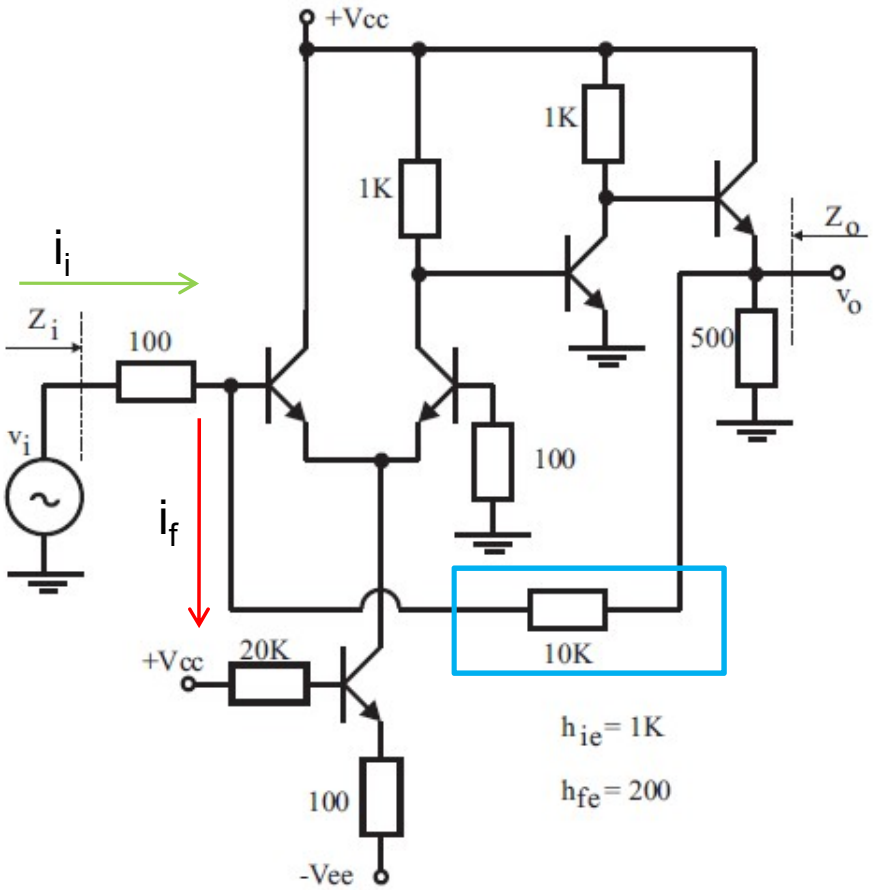


Figura 9: Ejercicio 10.

1- Determinar configuración

Paralelo - Paralelo

2 - Identificar red de realimentación

Parámetros [y]

3 – Ganancia requerida y adecuación de entrada de ser necesario

$$i_n = \frac{v_i}{100} \qquad A_z = \frac{v_o}{i_n}$$

Ejercicio 10 Guía 1

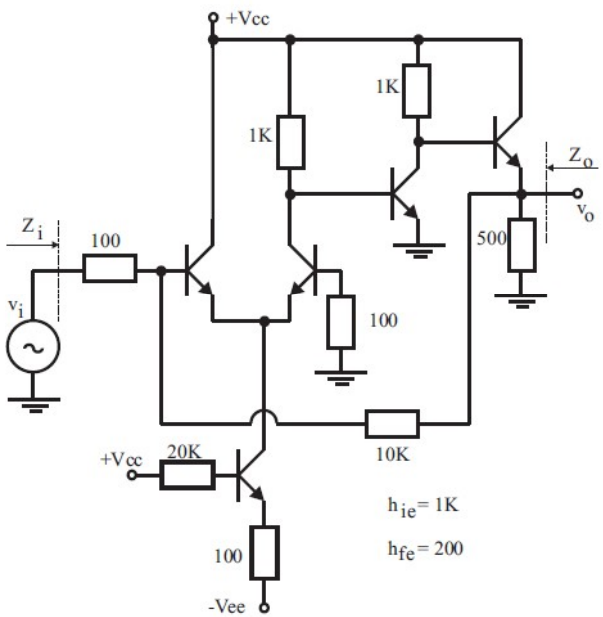
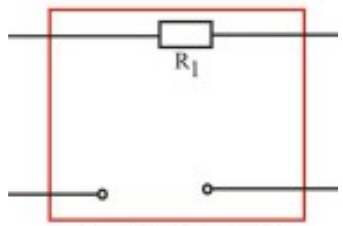
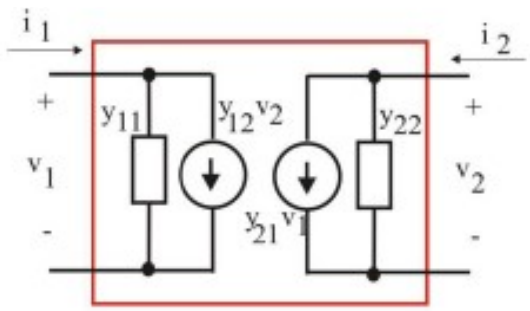


Figura 9: Ejercicio 10.



Red de realimentación



Red de realimentación

$$y_{11} = \left.\frac{i_1}{v_1}\right|_{v_2=0} = \frac{1}{R_1}$$

$$y_{12} = \left.\frac{i_1}{v_2}\right|_{v_1=0} = -\frac{1}{R_1}$$

$$y_{21} = \left.\frac{i_2}{v_1}\right|_{v_2=0} = -\frac{1}{R_1}$$

$$y_{22} = \left.\frac{i_2}{v_2}\right|_{v_1=0} = \frac{1}{R_1}$$

Ejercicio 10 Guía 1

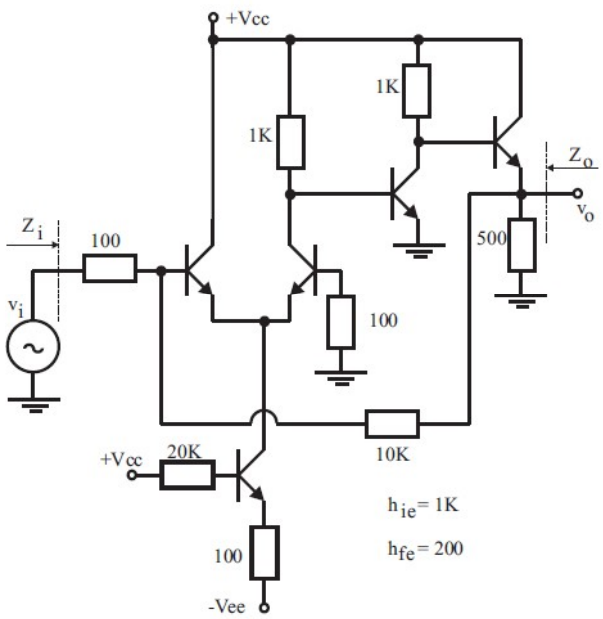
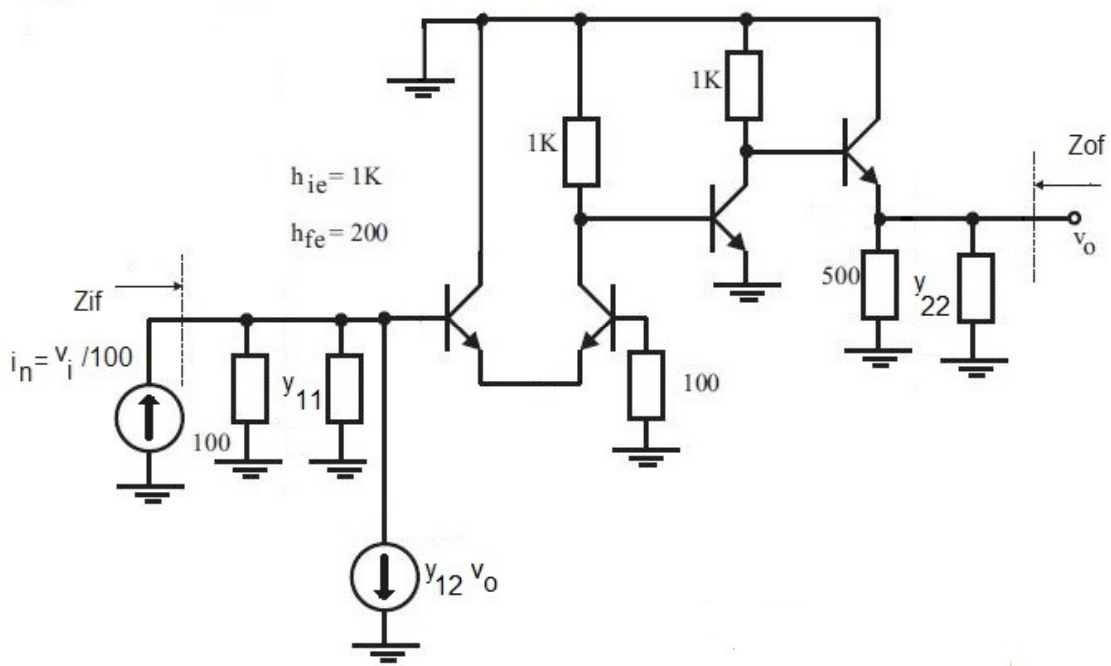
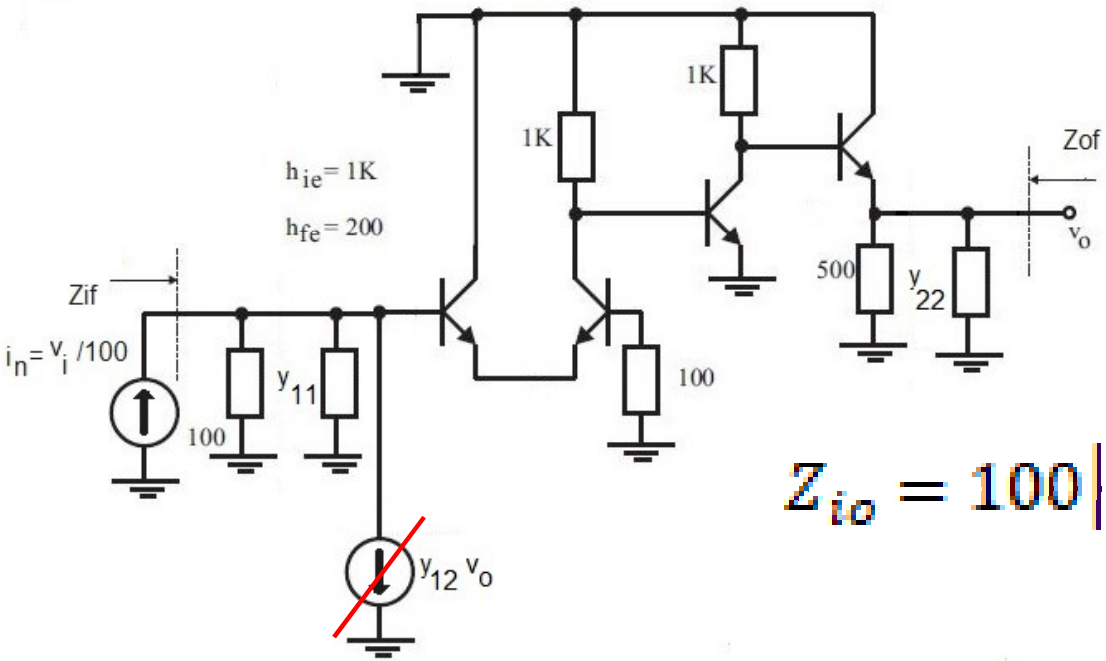


Figura 9: Ejercicio 10.



Ejercicio 10 Guía 1

Ganancia e impedancias sin realimentar con efectos de carga



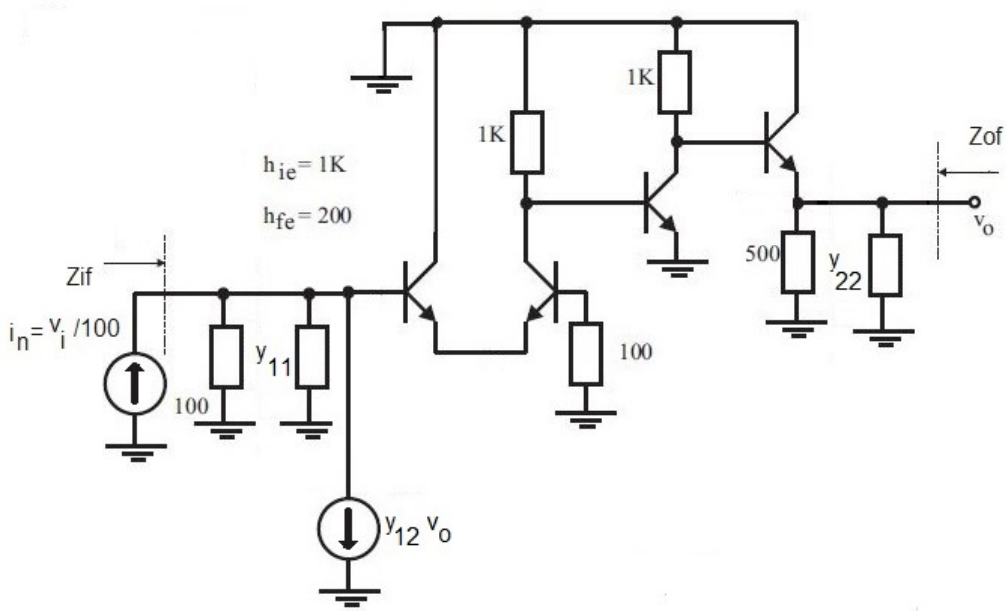
$$a_z = \frac{v_o}{i_n}$$

$$Z_{io} = 100 || 10K || (2h_{ie} + 100)$$

$$Z_{oo} = 500 || 10K || \frac{h_{ie} + 1K}{h_{fe} + 1}$$

Ejercicio 10 Guía 1

Ganancia e impedancias realimentadas



$$Con\ f = y_{12}\ y\ a_z$$

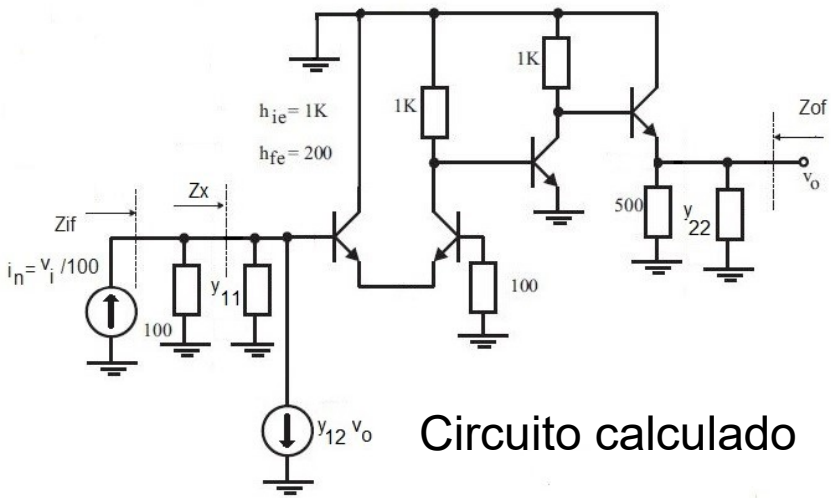
$$A_z = \frac{v_o}{i_n} = \frac{a_z}{1 + a_z f}$$

$$Z_{if} = \frac{Z_{io}}{1 + a_z f}$$

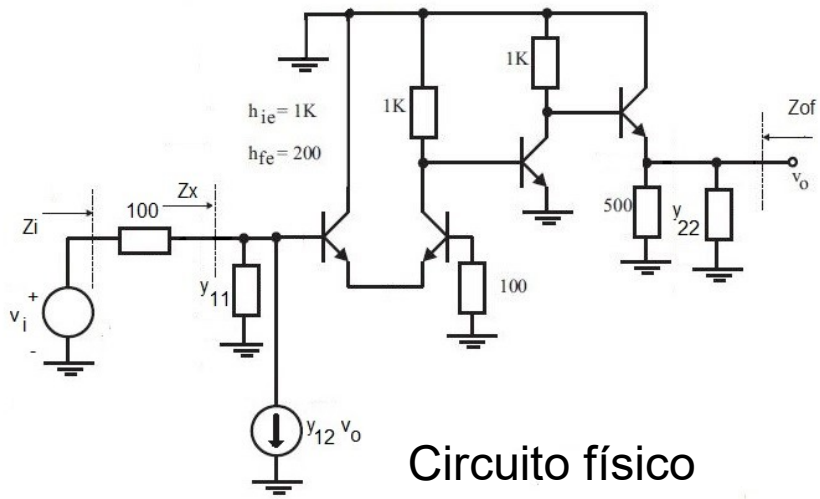
$$Z_{of} = \frac{Z_{oo}}{1 + a_z f}$$

Ejercicio 10 Guía 1

Adecuación a las transferencias e impedancias buscadas



Circuito calculado



Circuito físico

$$A_v = \frac{v_o}{v_i} = \frac{v_o}{i_n}\frac{i_n}{v_i} = A_z \frac{1}{100\Omega}$$

$$Z_o = Z_{of}$$

$$Z_{if} = 100\Omega || Z_x \quad \rightarrow \quad Z_i = 100\Omega + Z_x$$